

# Peiyang Xu

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## Education

### Electrical and Computer Engineering, Princeton University

Princeton, NJ

*Ph.D. student in Electrical and Computer Engineering*

Sep 2025 – Now

*Advisor: Prateek Mittal & Pramod Viswanath*

### Department of Computer Science and Technology, Tsinghua University

Beijing, China

*Bachelor of Engineering in Computer Science and Technology*

Sep 2021 – Jun 2025

- Courses: Software Engineering (A+), Introduction to Complex Analysis (A+), Student Research Training (A+), Modern Cryptography (A+), Computer Graphics (A), Artificial Neural Network (A)
- Research interest: AI Safety, Reinforcement Learning, Multi-Modal Models

## Research Experience

### Project: Contrastive RL for Enhanced Multimodal Grounding and Reasoning

Princeton, NJ

*Collaborator: Xingyu Fu Advisor: Prateek Mittal, Pramod Viswanath*

Dec 2025 – Present

- Identified a "coarse-grained" understanding bottleneck in current MLLMs, where models often over-rely on text reasoning and treat visual inputs as static, passive data.
- Curated a high-quality dataset of over 37,000 samples using a dual-path pipeline: Similarity-Based Filtering via image embeddings and AI-driven image editing (Nano Banana) to generate contrastive pairs.
- Developed a novel reinforcement learning objective by integrating a contrastive term into the GRPO algorithm, which enable the model to discern subtle visual features.
- Achieved consistent performance gains across both mathematical reasoning and general visual perception benchmarks.

### Project: Mitigating Object Hallucination in Large Vision-Language Models through Adversarial Contrastive Finetuning

Beijing, China

*Advisor: Xiaolin Hu, Associate Professor, Tsinghua University*

Feb 2025 – Aug 2025

- Explored the internal mechanism of object hallucination from the perspective of image embedding based on cosine similarity and Smooth Grad-CAM
- Proposed an adversarial contrast fine-tuning (ACFT) method to enhance the alignment between visual and textual embedding, thus mitigating object hallucination
- Provided a pipeline to automatically generating paired positive and negative examples using an adversarial hallucination attribute flipping (AHAF) method, followed by contrastive fine-tuning of the LVLM
- Achieved state-of-the-art performance on multiple benchmarks, e.g. outperforming existing approaches like VCD, OPERA and VTI, etc. on benchmarks like POPE and MME
- Contributed to a first-author paper submitted to ICML 2026

### Project: SafeVision: Efficient Image Guardrail with Robust Policy Adherence and Explainability

Chicago, IL

*Advisor: Bo Li, Associate Professor, Dept of CS, University of Chicago*

July 2024 – Nov 2024

- Proposed SafeVision, a novel image guardrail system that integrates human-like understanding and reasoning, with robust policy adherence, explainability, and lightning-fast inference speeds
- Provided VisionHARM-500K, a high-quality unsafe image benchmark comprising over 500k images to cover a wide array of risky categories, significantly broadening the scope and depth of unsafe image benchmarks
- Implemented an effective data collection and generation framework, a policy-following training pipeline, a customized loss function, and an efficient diverse QA generation strategy to enhance the training effectiveness
- Achieved state-of-the-art performance in both efficiency and accuracy, with an accuracy of 91.36% on VisionHARM-500K (17.36% higher than GPT-4o) and an inference time of 0.313 seconds per image

### Project: MMDT: Decoding the Trustworthiness and Safety of Multimodal Foundation Models

Beijing, China

*Advisor: Bo Li, Associate Professor, Dept of CS, University of Chicago*

Feb 2024 – Jun 2024

- Aimed to provide the first comprehensive safety and trustworthiness evaluation dataset and platform for multimodal foundation models (MMFMs), to help develop safer and more reliable MMFMs and systems
- Designed various evaluation scenarios/tasks and red teaming (attack) algorithms to generate challenging data and form a high-quality benchmark, to assess models from multiple perspectives, including safety, hallucination, fairness/bias, privacy, adversarial robustness, and out-of-distribution (OOD)
- Evaluated a range of state-of-the-art multimodal models, such as GPT-4o, Gemini, and Qwen, using our MMDT platform, and revealed several vulnerabilities and areas for improvement across all perspectives for these models
- Contributed to a paper accepted by ICLR 2025

**Project: Natural Language Induced Adversarial Images**

Beijing, China

Advisor: Xiaolin Hu, Associate Professor, Tsinghua University

July 2023 – April 2024

- Aimed to use prompt optimization to perform black-box attacks on commercial text-to-image models such as Midjourney and DALL-E-3, by generating natural adversarial examples in order to let classifiers make mistakes
- Utilized and improved non-differentiable optimization methods such as Genetic Algorithm and Ant Colony Algorithm to optimize the input prompt in order to achieve the attacking purpose
- Performed Genetic Algorithm in the discrete prompt space and assessed the fitness of individuals using a weighted result of attack success rate and image naturalness, with the naturalness measured by CLIP
- First to explore using prompt-based optimization strategies for generating adversarial images, with an attack success rate of 92.1% and superior in terms of naturalness, transitivity, and interpretability
- Analyzed adversarial images from a natural language perspective and demonstrated that the adversarial semantic information extracted from these images significantly affects the accuracy of traditional image classifiers
- Contributed to a co-first author paper accepted by the 2024 ACM Multimedia (ACM MM 2024)

**Project: CurBench: Curriculum Learning Benchmark**

Beijing, China

Advisor: Xin Wang, Assistant Researcher, Tsinghua University

Oct 2022 – Jan 2023

- Aimed to use curriculum learning to improve various classification algorithms and to establish the first benchmark for a systematic evaluation on curriculum learning
- Created a benchmark consisting of 16 datasets spanning 3 research domains (computer vision, natural language processing, and graph machine learning) under 3 settings (standard, noised, and imbalanced)
- Provided a unified pipeline that plugs automatic curricula into general machine learning process, supporting 14 core curriculum learning methods (personally responsible for the Meta Weight Net)
- Contributed to a paper accepted by the 2024 ICML

**Publications & Manuscripts**

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- **Peiyang Xu**, Xiaopei Zhu, Jun Zhu, Xiaolin Hu: Mitigating Object Hallucination in Large Vision-Language Models through Adversarial Contrastive Finetuning (submitted to ICML 2026, *under review*)
- **Peiyang Xu**, Minzhou Pan, Zhaorun Chen, Shuang Yang, Chaowei Xiao, Bo Li, [SafeVision: Efficient Image Guardrail with Robust Policy Adherence and Explainability](#)
- Chejian Xu, Jiawei Zhang, Zhaorun Chen, Chulin Xie, Mintong Kang, Zhuowen Yuan, Zidi Xiong, Chenhui Zhang, Lingzhi Yuan, Yi Zeng, **Peiyang Xu**, Chengquan Guo, Andy Zhou, Jeffrey Zhiwei Tan, Zhen Xiang, Zinan Lin, Dan Hendrycks, Dawn Song, Bo Li, [MMDT: Decoding the Trustworthiness and Safety of Multimodal Foundation Models](#) (ICLR 2025)
- Xiaopei Zhu\*, **Peiyang Xu**\*, Guanning Zeng, Yinpeng Dong, Xiaolin Hu, [Natural Language Induced Adversarial Images](#) (ACM MM 2024).
- Yuwei Zhou, Zirui Pan, Xin Wang, Hong Chen, Haoyang Li, Yanwen Huang, Zhixiao Xiong, Fangzhou Xiong, **Peiyang Xu**, Shengnan Liu and Wenwu Zhu, [CurBench: Curriculum Learning Benchmark](#), accepted by 42th International Conference on Machine Learning (ICML 2024)

**Honors & Awards**

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- Gordon Y.S. Wu Fellowship in Engineering, Princeton University(2025)
- Outstanding Graduate, Department of Computer Science and Technology, Tsinghua University(2025)
- Tsinghua University Research Excellence Scholarship (2024)
- Tsinghua University Academic Excellence Scholarship (2024)

**Skills & Others**

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**Language:** Chinese (Native); English (Fluent)**Exams:** TOFEL: 108 (Speaking 23); GRE: 158+170+4.0**Computer Science:** Proficient in Python, C++, and various other programming languages; Skilled in using deep learning libraries like PyTorch; Deep understanding of multimodal language models